Joint Frequency-Spatial Domain Network for Remote Sensing Optical Image Change Detection

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***Abstract*— Change detection for remote sensing images involves detecting regional surface changes of interest between two images taken of the same geographical area but at different times. In image processing, the spatial domain uses grayscale values to describe an image. The frequency is directly related to the spatial change rate, so the frequency domain can be intuitively associated with patterns of intensity variations in the image. These two domains provide different perspectives for image interpretation. Most existing deep-learning-based methods formulate change detection as a pixel-wise binary classification problem and utilize various strategies to extract information in the spatial domain. However, they rarely pay attention to the rich information in the frequency domain. To address this problem, we propose an end-to-end joint frequency-spatial domain network (JFSDNet) to implement remote sensing optical image change detection. Specifically, we introduce frequency information into the change detection to supplement the loss of image details caused by downsampling. In addition, we employ a frequency selection module to adaptively discriminate and choose frequency clues by reducing the complexity of the frequency features. The JFSDNet is applied to two publicly available datasets: the change-detection dataset (CDD) dataset and the LEarning VIsion and Remote sensing Change Detection (LEVIR-CD) dataset. Compared with other methods, both visual interpretation and quantitative assessment confirmed that our proposed method achieved a favorable performance.**

***Index Terms*— Change detection, deep learning, frequency domain, neural network, optical image.**

I. Introduction

CHANGE detection [1] is a technique for quantitatively analyzing and determining the characteristics and processes of feature changes from different temporal remote sensing images of the same geographical area. The change detection of high-resolution remote sensing images has been widely used in wide areas [2], such as land and resource management [3], [4], geological disaster monitoring [5], [6], urban expansion investigation [7], and coastal inundation mapping [8]. Nowadays, with the development of remote sensing technology, it is easy to obtain multitemporal high-resolution remote sensing data from many satellite sensors—for example, WorldView, QuickBird, and Gaofen-2—which form the basis of data to detect dynamic changes on Earth.

The general process of change detection is as follows: 1) obtaining the data and selecting valid data; 2) data preprocessing; 3) generating a difference image (DI) using specific algorithms; and 4) analysis of the DI to obtain a change map. Traditional change-detection techniques used handcrafted features to determine any changed regions or unchanged regions to generate DIs, such as change vector analysis (CVA) [9], principal component analysis (PCA) [10], and multivariate alteration detection (MAD) [11]. However, such methods have difficulty capturing high-level feature representations effectively, resulting in lower accuracy. Therefore, it is necessary to implement automatic change-detection techniques with strong representation capability. In recent years, deep learning has been explored in remote sensing, such as scene classification, pixel classification, ground object detection, and change detection of remote sensing images [12]–[15] owing to its advanced learning ability. Because of their outstanding capabilities in spatial domain feature learning, convolutional neural networks (CNNs) [16] are commonly used to extract spatial features for change detection [17]–[19]. Moreover, because of their ability to capture the temporal dependence between bitemporal data, recurrent neural networks (RNNs) are used to obtain features with spatial–temporal information. In addition, some researchers use architectures combining CNNs and RNNs to enhance temporal information between bitemporal images [20]–[23].

A. Motivation

The frequency domain refers to the analysis of mathematical functions or signals to frequency. As seen in current change-detection algorithm studies, images are primarily described in the spatial domain using grayscale values that convey global information. In contrast to images in the spatial domain, frequency-domain images are directly related to spatial rates

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of change, which focus on patterns of intensity variations in an image. Low frequencies correspond to the slowly varying grayscale components of an image, and higher frequencies correspond to more abrupt grayscale changes, which include the edges of objects and noise information [24]. Thus, the frequency and spatial domain information are complementary to each other. In [25] and [26], Lv *et al.* use the histogram to obtain the magnitude of changes and to analyze the trend of changes within the region of interest. Although these methods perform well, exploiting the potential synergy between frequency-domain analysis and a deep-learning-based approach remains an attractive research problem. During training, RGB images are gradually downsampled to obtain feature maps at different scales, but the detailed features are also gradually diluted. Consequently, it is extremely difficult to directly reconstruct a change map from the deep features. Therefore, it naturally occurred to us to involve frequency clues in deeply learned CNN models.

However, conventional frequency domains, such as the fast Fourier transform (FFT) and discrete cosine transform (DCT) [27], do not match the shift invariance and local consistency of natural images, thus simply transforming the original image to the frequency domain as a CNN input might not be feasible. Therefore, CNN-compatible frequency representation becomes pivotal if we want to leverage the discriminative representation power of learnable CNNs to extract related features for change detection.

B. New Contributions

Considering the above problems comprehensively, we propose a novel joint frequency-spatial domain network (JFSDNet) for remote-sensing optical image change detection. Our framework addresses two problems:

1) How to explicitly employ frequency-domain information in remote sensing optical image change detection to complement the spatial domain detailed information lost during downsampling; and
2) How to adaptively select more relevant frequency-related features?

First, the spatial data are converted to the frequency domain using a single preprocessing branch. Subsequently, the frequency selection module (FSM) is exploited to choose different frequency components which can supplement mid-level and high-level features. Next, the data in the two domains are fed into a hierarchical network for feature extraction and reconstruction. Consequently, our proposed JFSDNet achieves unification of the frequency and spatial domains.

Contributions in this article can be summarized as follows:

1) A novel jointly frequency-spatial domain method is proposed for remote sensing image change detection. In view of lacking the full use of the frequency information in existing deep learning-based change-detection methods, we propose the JFSDNet to explicitly exploit the frequency responses to complement spatial information of the mid-level and high-level features in the neural network. The proposed model preserves the correlation between adjacent pixels in natural images, thus mining through CNNs effectively.
2) We design an FSM, which adaptively selects frequencies relevant to image details. FSM can select partial frequencies to reduce the complexity of the frequency information, so that frequency features fed into the network are as relevant as possible to the needs of the network.
3) Extensive experiments are conducted to reveal the robustness and generalization of the proposed architecture. Compared with other methods, the proposed JFSDNet could significantly boost the performance of capturing changes in high-resolution remote sensing images.

The remainder of this article is organized as follows: Section II discusses the related works of this study. Section III elaborates on the JFSDNet. Section IV presents the experimental settings, and the experimental results for real remote sensing datasets are presented in Section V. Section VI concludes this article.

II. Related Work

Existing change-detection methods for remote sensing optical images can be divided into traditional and deep-learning-based methods.

A. Change Detection

1) Traditional Methods: Traditional change-detection methods can be divided into algebraic-operations-based, transformation-based, and classification-based methods. The most common method is to use algebraic operations to generate a DI for bitemporal data and use threshold segmentation or other strategies to obtain the change map. CVA separates the changed pixels from unchanged pixels by binarizing bitemporal DIs. Du *et al.* [28] introduce an additional image and generate preliminary change results using CVA for logic verification. Methods based on image transformation can learn specific feature representations from the original image. Celik *et al.* [10] employ PCA to identify the major components among numerous features and extract eigenvectors from image blocks for unsupervised change detection. MAD [29] is an algorithm that explores canonical correlation analysis to find linear combinations to maximize spectral variations. The most common classification-based approach is postclassification comparison (PCC). PCC first classifies the data at various phases in the observation and then compares and analyzes the differences between each phase, to arrive at the final multitemporal change detection. For instance, Walter [30] classifies the pixel groups in a geographical information system based on the maximum likelihood classification and then obtains land cover change information by comparison.

Although easy to implement, the traditional change-detection methods have a limited capability to capture complex visual features. As the amount of remote sensing image data continues to increase, it remains difficult for traditional methods to satisfy the change detection for large-scale, fine-grained remote sensing optical images.

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2) Deep Learning-Based Methods: With the development of deep learning in the field of computer vision, neural-network-based change-detection methods [19], [31]–[37] have become a hot research topic and have made remarkable progress. Because of the superior performance of CNNs in adaptively extracting features, many researchers have utilized CNN-based change-detection models to achieve higher accuracy rates than other traditional methods. Zhan *et al.* [38] feed bitemporal images into a five-layer Siamese network trained with a weighted contrastive loss to extract the spatial features and then generate the final change maps using k-nearest-neighbor techniques.

Several change-detection methods use architectures with powerful feature extraction capabilities pretrained on datasets as a baseline, for example, Visual Geometry Group Net (VGG-Net) [39] and Resnet [40]. Zhang *et al.* [41] use pretrained layers before the pool5 layer of the VGG16 as a backbone to extract spatial features at different levels. Shi *et al.* [31] construct a feature extractor based on Resnet-18 to learn the spatial information in bitemporal images and then compare the change map by measuring the embedding eigenvectors. Chen *et al.* [42] choose Resnet50 as the basic network and utilize spatial and channel attention modules to improve change feature discrimination.

Researchers have improved the fully convolutional (FC) network [43] based on Unet [44] and applied it to change detection, achieving outstanding efficiency. Daudt *et al.* [33] develop two Siamese extensions of Unet—FC-Siam-conc and FC-Siam-diff—which extract and fuse the spatial features of each time phase by concatenation and difference, respectively. Liu *et al.* [8] extract the bitemporal and dual-polarization features from synthetic aperture radar (SAR) images using an improved Unet to detect the changes in a floodplain. Peng *et al.* [34] use the Unet++ network with dense nodes to extract different levels of spatial features from bitemporal data to capture subtle changes in remote sensing optical images. Zheng *et al.* [35] design an Unet-like backbone that fused multiscale spatial–temporal features and multilevel context information through cross-layer blocks.

The RNN handles temporal data through a recurrent hidden state with activation at each time step, depending on past computations. It can address the temporal connections between multitemporal sequence data in change-detection tasks. Lyu *et al.* [36] employ an improved long short-term memory (LSTM) model to learn temporal features from long-term sequence remote sensing optical data. To exploit spatial–temporal information, several methods have combined CNNs and RNNs. Mou and Zhu [21] propose an end-to-end model combining an RNN and CNN—named an ReCNN—to generate a rich joint spectral–spatial temporal-feature representation from multispectral images. Liu *et al.* [22] combine a CNN with bidirectional LSTM to extract spectral–spatial features and the temporal dependence of bitemporal images to improve the detection performance. Chen *et al.* [37] design stacked LSTM units to map spatial–spectral features extracted by a CNN into a new feature space and mine latent temporal change information.

However, the above methods focus on extracting spatial and temporal domain information; they do not explicitly consider frequency-domain characteristics. Our proposed method introduces the frequency coefficients into an end-to-end change-detection network.

B. Frequency-Domain Study Based on Deep Learning

Frequency-domain analysis is a classical and significant method in image signal processing and has been widely applied in several applications such as image classification [45], [46], face forgery detection [47]–[49], and image reconstruction [50], [51]. Gueguen *et al.* [46] decode Joint Photographic Experts Group (JPEG) images partially and train CNNs directly on DCT coefficients for faster and more accurate processing. Liu *et al.* [50] embed wavelet transform into CNN architecture as a downsampling operation to reduce spatial resolution, enlarge the receptive field, and then deploy inverse wavelet transform to reconstruct feature maps. Mathieu *et al.* [45] compute convolution as products in the Fourier domain, which accelerates training and inference using convolutional networks. Wang *et al.* [48] separate the frequency domain into different bands with three filters and then invert them back into the RGB domain to extract features. The obtained feature is combined with RGB features for deep fake detection.

III. JFSDNet for Change Detection

In this section, we propose an end-to-end network that introduces joint features in the frequency domain as a complement of spatial features to capture fine-grained change information. A flowchart of the entire network is shown in Fig. 1. The architecture is divided into four parts: frequency-domain transform, frequency selection, hierarchical change-detection reconstruction, and change map generation. The input of the model includes two components, that is, spatial- and frequency-domain images. The frequency-domain data are selected by the FSM and then fed into the neural network with the spatial domain data. These two kinds of features are used as low-level and high-level feature inputs to the reconstruction network, respectively. A nested encoding–decoding hierarchical network with multiple levels is used to extract the features and generate change maps. The outputs of multiple levels are embedded in the final output, which can capture fine-grained changes.

A. Frequency-Domain Transform and Selection

1) Frequency-Domain Transform: The input of the JFSDNet comprises two components, that is, spatial-domain images and frequency-domain data. Fig. 2 shows the data preprocessing based on the frequency-domain transform.

Inspired by JPEG [52], we apply a partial process of JPEG encoding to the transformation of bitemporal frequency-domain data. However, different from it, the spatial resolution of the chroma channels is not reduced in our method, so we do not need to consider the compatibility of different channels. Without loss of generality, let $X_1 \in \mathbb{R}^{H\times W\times 3}$ denote the input RGB image, where H and W denote the height and width of

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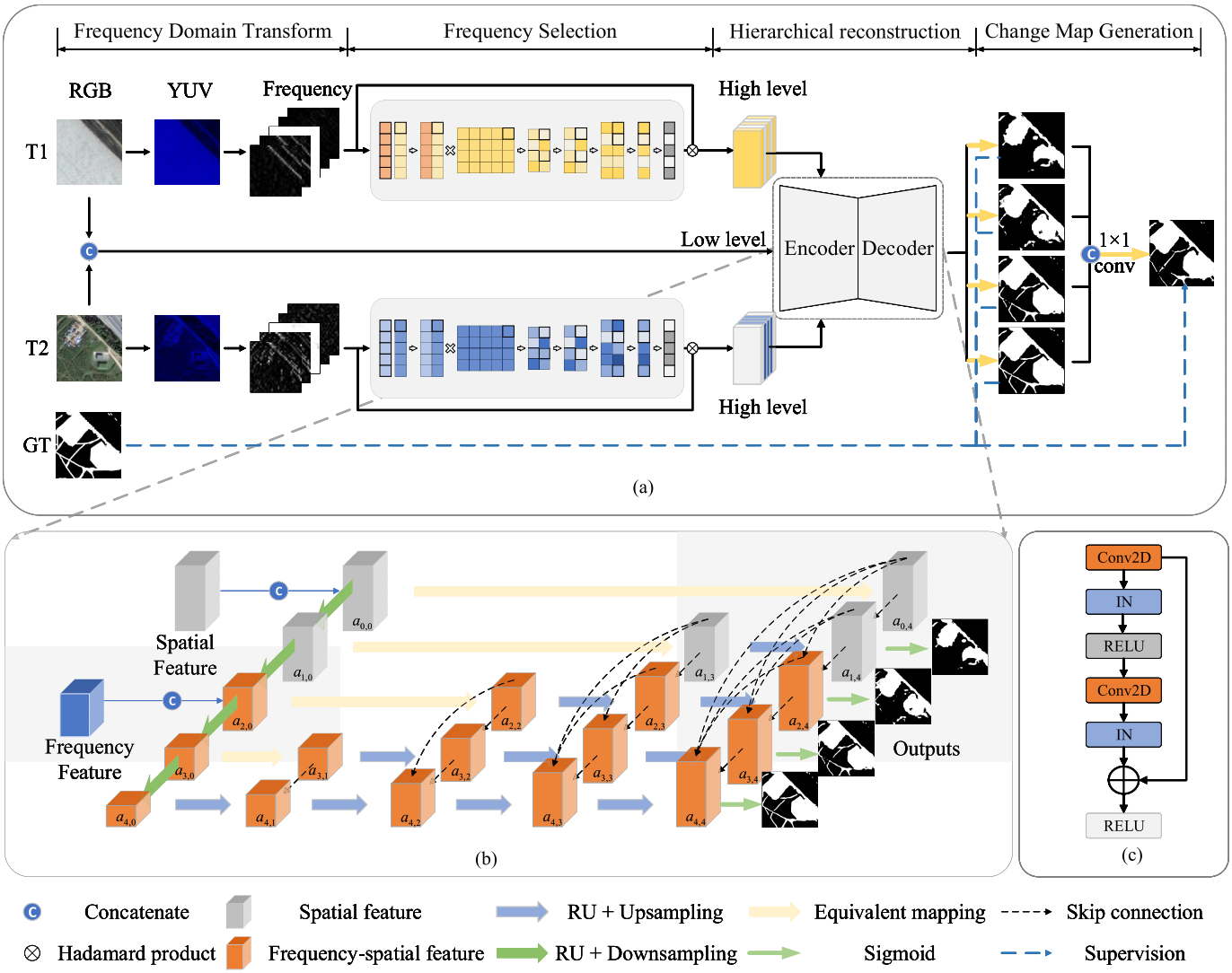


Fig. 1. (a) Architecture of the proposed JFSDNet, which is divided into four parts: frequency-domain transform, frequency selection, hierarchical change-detection reconstruction, and change map generation. (b) Hierarchical change-detection reconstruction. Spatial-domain features and frequency-domain features are used as low-level and high-level feature inputs to the reconstruction network, respectively. (c) RU and IN denote instance normalization.

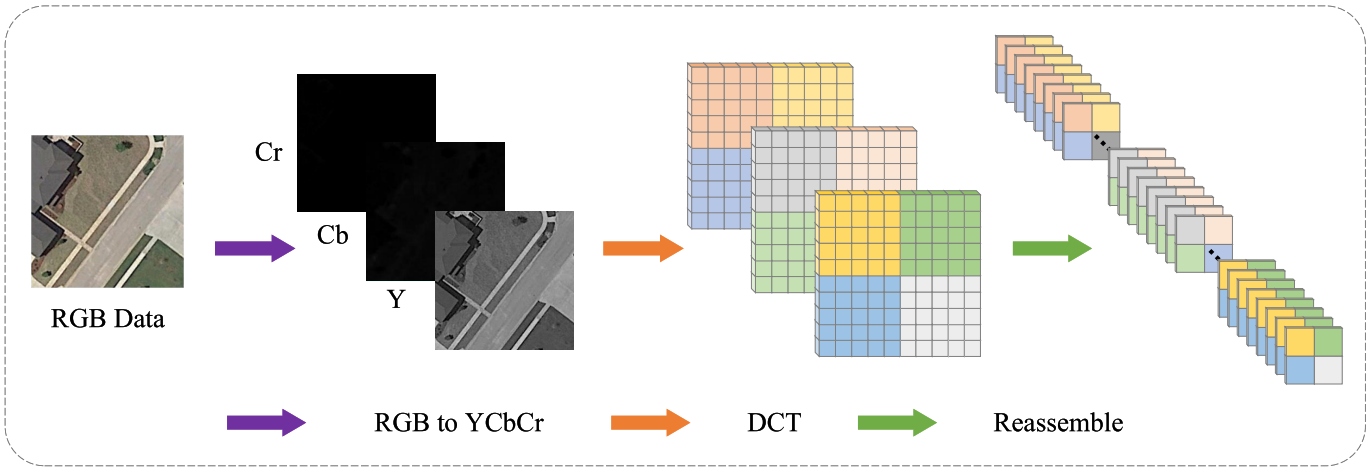


Fig. 2. Flow of the frequency-domain transformation. The bitemporal data are converted into frequency-domain data by this process.

the image, respectively. We first transform X_1 from the RGB space to the YCbCr space and then reassemble it into the frequency-domain data after the DCT. This processing can be expressed as follows:

$$X_2 = \mathcal{R}[D(AX_1 + B, b)] \tag{1}$$

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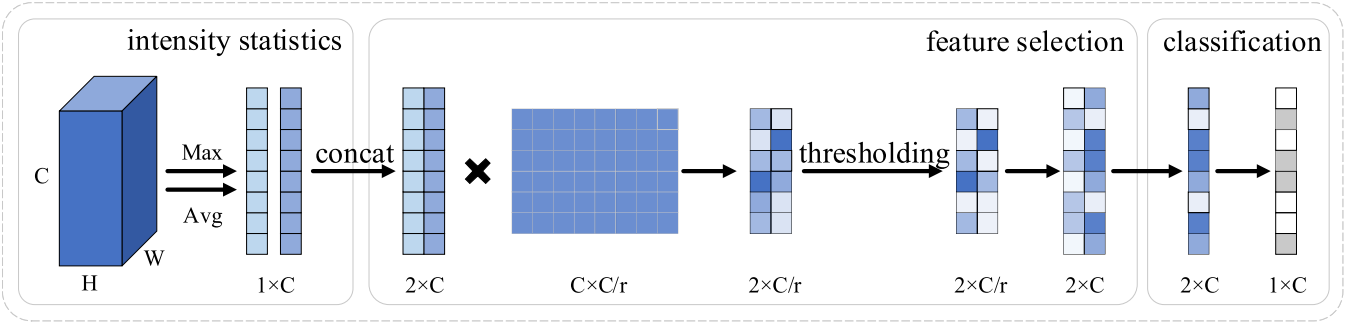


Fig. 3. FSM. Different colors denote the intensity statistics.

where A and B are the reference matrices from [52], D denotes the 2-D-DCT [27], which has a compact layout of frequency distribution, that is, most energy is concentrated on the low frequency and is placed in the top-left corner, and the other is in the high-frequency region located in the bottom-right corner. b denotes the size of the DCT blocks, which is set to 8. $\mathcal{R}$ denotes the operation of reassembling the same frequency coefficient.

2) Frequency Selection Module: After the operations in Section III-A1, the input data, $X_1 \in \mathbb{R}^{H\times W\times 3}$, is converted to a frequency response, $X_2 \in \mathbb{R}^{H/8\times W/8\times 192}$. Because different frequencies play different roles in the image structure, we speculate that not all coefficients are valid for change detection. Irrelevant information contributes to increasing the network training difficulty.

Consequently, to make the network more effective, we design an FSM that can adaptively select frequency features. The details of the FSM are presented in Fig. 3.

First, we perform intensity statistics on the frequency-domain data X_2 obtained in Section III-A1. Since X_2 is the reassembled data at different frequencies in each channel, we can obtain the global and local response at different frequencies by analyzing the average and maximum intensity of the channels, as calculated by

$$\alpha^1 = \text{Avg}(X_2) = \frac{1}{h\times w}\sum_{i=1}^{h}\sum_{j=1}^{w} x_2(i,j) \tag{2}$$

$$\alpha^2 = \text{Max}(X_2) = \text{Max}(x_2(i,j)). \tag{3}$$

Then we concatenate α^1 and α^2 as tensor X_3, of shape $2\times C$ (C denotes the channel number of input, which is equal to 192). Considering the redundant part of the frequency information, we perform a matrix multiplication of X_3 to linearly reduce its dimensionality, which is

$$X_4 = W_d^T X_3^T \tag{4}$$

where matrix $W_d \in \mathbb{R}^{C\times C/r}$ is randomly initialized and learnable. r denotes a reduction ratio, which is set to 16. To reduce the complexity of the features, we use a threshold function $f(\cdot)$ that binarizes the intensity responses to represent the different importance, which is

$$X_5 = f(X_4) \tag{5}$$

$$f(x) = \begin{cases} 0, & x \le 0 \\ x, & x > 0. \end{cases} \tag{6}$$

Here, $f(\cdot)$ is similar to Rectified Linear Units (RELU), which is used to reduce computational complexity, while improving the data sparseness after the matrix scale transformation [53]. Thereafter, we revert the tensor X_5 to the original frequency channel dimension. Thus, the data matrix is resized to $2\times C$. Finally, we implement the selection of different frequencies by a linear classifier, $\mathcal{G}$, which is

$$X_6 = \mathcal{G}\left(1_{1\times 2}\left(W_u^T X_5\right)^T\right) \tag{7}$$

where $W_u \in \mathbb{R}^{C/r\times C}$. In the inference phase, the values in X_6 represent the probability of selecting the frequency. However, random sampling is not differentiable in the back-propagation process. Therefore, we use the Gumbel-SoftMax reparameterization technique [54] to determine how to choose different frequency characteristics, as follows:

$$y_i = \frac{\exp((\log\pi_i + g_i)/\tau)}{\sum_{j=1}^{k}\exp\left((\log\pi_j + g_j)/\tau\right)}, \quad \text{for } i = 1,\ldots,k \tag{8}$$

where π_i denotes the probability of each category, and $g_{1,\ldots,k}$ is an i.i.d. sample drawn from the sequence of standard Gumbel random variables. τ is the temperature. Subsequently, we obtain a set of coefficients of size C that are either 0 or 1, where 0 and 1 represent whether that frequency channel is selected or not.

In short, the final feature map can be obtained through the FSM and is computed as follows:

$$Y = x \otimes \mathcal{G}\left\{1_{1\times 2}\left[W_u^T f\left(W_d^T[\text{Ave}(x);\text{Max}(x)]^T\right)\right]^T\right\} \tag{9}$$

where $\mathcal{G}$ is the Gumbel-SoftMax technique, $W_d \in \mathbb{R}^{C/r\times C}$, $W_u \in \mathbb{R}^{C\times C/r}$ denotes the weights of matrixes for changing the scale of the tensor, [;] is the concatenate operation in the matrix, and $\otimes$ denotes the Hadamard product. From formula 9, it is concluded that the final frequency-domain feature Y retains paramount frequencies and selectively removes redundant frequencies. The FSM enhances the identifiability of the frequency and highlights the feature representation of the region of change.

B. Hierarchical Change-Detection Reconstruction Network

The hierarchical change-detection reconstruction network is built on a deep-supervised end-to-end network and can be regarded as the nesting of four scales of separate encoder–decoder networks, as shown in Fig. 1(b). These four networks use the same encoder but reconstruct them at

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different levels. Specifically, the JFSDNet uses residual blocks (RBs), as shown in Fig. 1(c), as the basic unit to extract features. We write down the residual unit (RU) in the upper left corner as $a_{0,0}$, the lower corners of the other positions gradually increasing.

Multiscale feature representation can enlarge receptive fields for better parsing scenes, while multilevel feature representation can aggregate semantic context information and spatial details. Therefore, we use different ways to connect multiscale and multilevel information. In the vertical direction, downsampling is used to link $a_{i,0}$ to each other. As i increases, the feature map size steadily decreases. The shallow features are weak for semantic information but excellent for geometric information, whereas deeper features are the reverse. To balance the contradiction between them, we use the frequency-domain data produced in Section III-A1, which has the same size as the low-resolution feature map $a_{3,0}$, as the input to efficiently employ the frequency information and provide more relevant and effective supplementary information for the current resolution in the spatial domain. These frequency data preserve the original image details and perform spatial consistency and translation invariance for spatial images.

In the spatial domain, the RGB images are concatenated and fed into the network to extract spatial-domain features. In the frequency domain, we add multiscale information to the raw image twice the size.

In the horizontal direction, the features of different levels are related to one another in a variety of ways: the jth feature in each row can be obtained by upsampling the $j-1$th feature, as follows:

$$a'_{i,j} = \begin{cases} a_{i,j}, & j = 4 - i \\ a'_{i,j-1}, & j > 4 - i. \end{cases} \tag{10}$$

The features in each column are densely connected and can be formulated as follows:

$$a'_{i,j} = a_{4-j,j} \oplus a_{4-j+1,j} \oplus \cdots \oplus a_{i,j}, \quad i \geq 4 - j \tag{11}$$

where $i \in \{0, 1, 2, 3, 4\}$ and $j \in \{1, 2, 3, 4\}$, $a'_{i,j}$ denotes the feature of the input next level RU, and $\oplus$ denotes concatenation.

Based on the above, the hierarchical reconstruction network can be regarded as a process of reconstructing two feature streams, that is, spatial features and joint frequency-spatial feature reconstruction [shown in the gray and orange blocks in Fig. 1(b)]. When $i \geq 2$, the joint frequency-spatial features in these three deeper-level networks play a significant role in reconstruction and can produce high-quality change maps.

C. Change Map Generation

According to Fig. 1(b), the outputs of the encoder–decoder are four change maps from different levels. The four change maps are on the same scale. These four outputs represent different levels of information that can be combined to learn more meaningful features [55], ranging from low to high levels. We first concatenate the four change maps in the channel dimension. Then we use a 1×1 convolution with two kernels to extract the feature of change maps. After that, a sigmoid layer is used for pixel-level binary classification. We denote these four side outputs as $A = \{a_{4,j}, j = 0, 1, 2, 3\}$. The fusion of the multilevel outputs can be expressed as follows:

$$\hat{Y} = \sigma\left(\text{conv}_{1\times1}\left(a_{4,0} \oplus a_{4,1} \oplus a_{4,2} \oplus a_{4,3}\right)\right) \tag{12}$$

where $\hat{Y}$ denotes the fusion output, $\oplus$ denotes concatenation, $\text{conv}_{1\times1}$ is a 1×1 convolution layer, and σ denotes the sigmoid function.

D. Loss Function

As the change detection is considered a binary classification task of "change" and "nonchange" for each pixel, we select weighted binary cross-entropy (WBCE) loss with a class-balancing parameter, β, as our loss function in this article. Specifically, we define the class-balanced binary cross-entropy loss function as follows:

$$L_{\text{WBCE}}(X, Y) = -\frac{1}{N}\sum_{i=1}^{N}\left[(1-\beta)(1-y_i)\cdot\log(1-x_i) + \beta y_i \cdot \log x\right] \tag{13}$$

where $X = (x_i, i = 1, 2, \ldots, N)$ and $Y = (y_i, i = 1, 2, \ldots, N)$, $y_i \in \{0, 1\}$ denote the output image and the corresponding ground truth, respectively. If $y_i = 0$, the ground-truth value belongs to the unchanged class; otherwise, $y_i = 1$. x_i represents the probability that the pixel is predicted as a changed class.

To maximize the role of different levels of output in the objective function, we denote the corresponding weight to be $w = (w_0, w_1, w_2, w_3)$. We combine the multiside outputs and the final change map for supervision. In particular, our object function can be expressed as

$$L_{\text{total}} = w_0\text{Dis}(Y, a_{0,4}) + w_1\text{Dis}(Y, a_{1,4}) + w_2\text{Dis}(Y, a_{2,4}) + w_3\text{Dis}(Y, a_{3,4}) + \lambda Dis\left(Y, \hat{Y}\right) \tag{14}$$

where $\text{Dis}(\cdot, \cdot)$ is the distance between the fused predictions and the ground truth, which we set as the WBCE loss. λ denotes the weight of the fusion output. After fine-tuning the validation split, we set the weights to $\lambda = 1$ and $w = \{0.5, 0.5, 0.75, 0.75\}$ in our experiments to ensure the weight of the joint frequency-spatial feature in the network.

IV. Experimental Setting

A. Dataset

1) Change-Detection Dataset: The change-detection dataset (CDD) [56] released by Lebedev contains season-varying optical satellite images of the same region from Google Earth. The dataset consists of 16 000 images clipped by 256×256-pixel patches from original real-time seasonal images, distributed with 10 000 training sets, 3000 test sets, and 2998 validation sets. The ratio between the samples of "changed" and "unchanged" areas is about 1:7. It should be noted that Lebedev only considers changes due to the disappearance and appearance of objects and ignores changes due to factors such as seasonal differences, as shown in Fig. 4(a).

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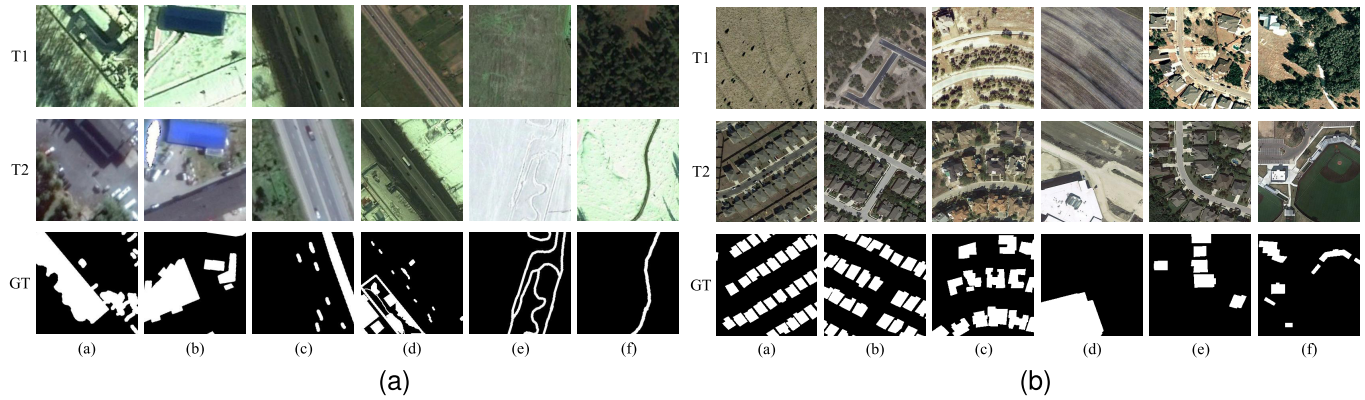


Fig. 4. Dataset (a): Several samples (256 × 256) of the different change types in the CDD dataset. (a) and (b) Building change. (c) and (d) Minor object change. (e) and (f) Road change. Dataset (b): Several cropped samples (256 × 256 pixels) of the different change types in the LEVIR-CD dataset. The samples show the change in different types of buildings in the dataset. The first and second rows indicate the bitemporal image pairs. The third row indicates the ground truth.

2) LEVIR-CD Dataset: The LEarning VIsion and Remote sensing Change Detection (LEVIR-CD) dataset consists of 637 image pairs of size 1024 × 1024 pixels, divided into training, validation, and test sets. These bitemporal images are obtained from 20 different regions, collected from Google Earth. Based on [32], we cut images into 256 × 256-pixel patches in the training and test. Therefore, we obtain 7120 training sets, 1024 validation sets, and 2048 test sets. The ratio between the samples of "changed" and "unchanged" areas is about 1:21. The challenges of this dataset include the uneven distribution of samples and building change detection, as shown in Fig. 4(b).

B. Comparative Methods

To assess the efficacy of the proposed method, we compare it to the following state-of-the-art methodologies, which are listed below.

1) FC-EF: FC-EF [33] uses an early fusion (EF) strategy to concatenate the input of the network. The features are transported from the encoder to the decoder using skip connections, thereby producing change maps with complete boundaries.

2) FC-Siam-Conc: FC-Siam-Conc [33] is a Siamese extension of Unet. The encoder uses a weight-sharing fully convolutional network to extract features from bitemporal images, which are subsequently concatenated in the decoder.

3) FC-Siam-Diff: FC-Siam-Diff [33] is like FC-Siam-Conc. The encoder extracts features using a weight-sharing network. The difference is that the absolute value of the feature differences is concatenated in the decoder to skip the connection.

4) FCN-PP: FCN-PP [6] is a fully convolutional network with a pyramidal pool (FCN-PP). The network captures a wider receptive field and is used for landslide detection in remote sensing images.

5) Unet++ With MSOF: Unet++ with multiple side-output fusion (MSOF) [34] uses early fused bitemporal data as its input, incorporates multilevel information, and performs deep supervision.

6) Deeply Supervised Image Fusion Network (DSIFN): DSIFN [41] is based on late fusion and uses the VGG-Net to extract features at different levels in the encoder. It uses an attention mechanism and deep supervision of features at different levels in the decoder.

7) Bitemporal Image Transformer Change Detection (BIT-CD): BIT-CD [57] introduces a bitemporal image transformer in a deep feature differencing-based framework to model contexts within the spatial–temporal domain.

8) Dual-Domain Network (DDNet): DDNet [58] introduces the DCT domain to alleviate the speckle noise in the SAR images to extract the change features accurately.

C. Implementation Details

The experiments are implemented on a PyTorch (version 1.0.1) platform using a 16-core Advanced Micro Devices (AMD) Ryzen Threadripper 2950× workstation with an NVIDIA GTX 1080ti graphic processing unit (GPU). The batch size is 16 and the initial learning rate is 1×10^{-3} for the two datasets. The networks are optimized using the Adam optimization algorithm. The kernel size of the convolutional layers is set to 3. To reduce the overfitting effect and improve the generalization ability of the proposed network, we use random data augmentation operations in the training process, including a random rotation at $n \times 90°(n = 0, 1, 2, 3)$, flipped in the horizontal and vertical directions.

D. Evaluation Metrics

The change-detection task can be thought of as a binary classification project for change and nonchange. We use precision (Pre), recall (Rec), F1-score (F1), overall accuracy (OA), total error (TE), average accuracy (AA), false alarm (FA), missed alarm (MA), and Kappa coefficient (KA) as evaluation metrics to assess the performance of the proposed method. Given that TP, FP, TN, and FN refer to true positive, false positive, true negative, and false negative, respectively, Pre, Rec, F1, OA, TE, AA, FA, MA, and KA can be described as follows:

$$\text{Pre} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{15}$$

$$\text{Rec} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{16}$$

$$\text{F1} = \frac{2\text{Pre} \times \text{Rec}}{\text{Pre} + \text{Rec}} \tag{17}$$

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TABLE I
RESULTS OF DIFFERENT APPROACHES ON THE CDD DATASET

Method	Pre	Rec	F1	OA	TE	AA	MA	FA	KA
DDNet (2022, [58])	0.4942	**0.9314**	0.3010	0.7624	0.2376	0.6591	0.7820	0.0615	0.1960
FC-EF (2018, [33])	0.8677	0.5435	0.6616	0.9359	0.0641	0.9064	0.4356	0.0117	0.6283
FC-Siam-Conc (2018, [33])	0.8524	0.6189	0.7137	0.9426	0.0574	0.9058	0.3619	0.0145	0.6825
FC-Siam-Diff (2018, [33])	0.8852	0.5871	0.6996	0.9427	0.0573	0.9197	0.3904	0.0104	0.6694
FCN-PP (2019, [6])	0.9267	0.8024	0.8586	0.9717	0.0283	0.9565	0.1735	0.0078	0.8426
Unet++MSOF (2019, [34])	0.8954	0.8711	0.8756	0.9673	0.0327	0.9663	0.0941	0.0073	0.9076
DSIFN (2020, [41])	0.9496	0.8608	0.9030	0.9771	0.0229	0.9650	0.1390	0.0064	0.8902
BIT-CD (2021, [57])	0.9414	0.9224	0.9316	0.9741	0.0259	0.9639	0.2009	0.0052	0.8540
JFSDNet	**0.9875**	0.9260	**0.9555**	**0.9907**	**0.0093**	**0.9899**	**0.0653**	**0.0015**	**0.9557**

$$\mathrm{OA} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}} \tag{18}$$

$$\mathrm{TE} = \frac{\mathrm{FP}+\mathrm{FN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}} \tag{19}$$

$$\mathrm{AA} = = \frac{1}{2} \times \left(\frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} + \frac{\mathrm{TN}}{\mathrm{TN}+\mathrm{FN}} \right) \tag{20}$$

$$\mathrm{FA} = \frac{\mathrm{FP}}{\mathrm{FP}+\mathrm{TN}} \tag{21}$$

$$\mathrm{MA} = \frac{\mathrm{FN}}{\mathrm{TP}+\mathrm{FN}} \tag{22}$$

$$P_e = \frac{(\mathrm{TP}+\mathrm{FP}) \times (\mathrm{TP}+\mathrm{FN}) + (\mathrm{FN}+\mathrm{TN}) \times (\mathrm{FP}+\mathrm{TN})}{(\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN})^2} \tag{23}$$

$$\mathrm{KA} = \frac{OA - P_e}{1 - P_e}. \tag{24}$$

V. RESULTS AND DISCUSSION

In this section, to prove the effectiveness of the JFSDNet, we compare it with other deep-learning-based change-detection methods, as described in Section IV-B.

A. Comparisons on the CDD Dataset

1) Quantitative Evaluation: Table I lists the change-detection results for the CDD dataset. The best scores are highlighted in bold, and the second-best scores are underlined. The JFSDNet achieves the best Pre, Rec, F1, OA, and TE, reaching 0.9875, 0.9260, 0.9555, 0.9907, and 0.0093, respectively. Compared to the BIT-CD, the JFSDNet increased the F1 score by 0.0239, and the OA by 0.0166. The aforementioned quantitative results illustrate the proposed JFSDNet's excellent performance and effectiveness in remote sensing change detection. As the results show, the proposed JFSDNet could extract spatial–frequency-domain features, mine change information exhaustively from high-resolution images.

2) Visual Performance: Four classic scenes of the CDD dataset were chosen for evaluation among the acquired findings to qualitatively evaluate the effectiveness of the JFSDNet: road changes, minor object changes, building changes, and complex changes. The results are shown in Fig. 5.

The reason for the higher performance of FC-Siam-Conc and FC-Siam-Diff in detecting most sorts of variations, especially building changes, is that the Siamese structure kept the features in the bitemporal images. FCN-PP employs a pyramid structure and pretrained models to obtain better performance. Due to its dense connectivity and multifarious output, Unet++MSOF can detect nearly all forms of changes, except for snow-covered roads [as shown in Fig. 5(h)]. DSIFN is excellent for most change types due to its supervision at different depths, the attention mechanisms suppressing irrelevant features across channels and spatial dimensions. Nevertheless, the change maps generated by DSIFN have blurred edges, especially on small objects. Individual objects become difficult to discern [see the green and red regions in Fig. 5(i)]. BIT-CD uses a transformer to learn contextual information and has achieved pretty good results in various scenarios.

The best performance is attained with JFSDNet, as can be observed in Fig. 5(k). In comparison to existing methods, the proposed methodology is capable of constructing change maps with more accurate boundaries, even if the objects are covered by snow. Regarding the change detection of elusive changes, JFSDNet can detect almost all the appearance and disappearance of minor objects. Despite the fact that the JFSDNet's detected borders are smaller than the ground truth, the dense objects can be distinguished from one another, whereas other approaches tend to identify as a single change or miss detection. The change-detection results of buildings and complicated scenes are shown in Fig. 5(j), and the proposed JFSDNet likewise has the best visual performance. The JFSDNet can extract and retain the borders and forms of tiny changes more efficiently, as seen in Fig. 5. In particular, it can overcome the interference of seasonal changes, exhibiting strong robustness.

3) Reason: The aforementioned comparisons illustrate the excellent performance and effectiveness of our proposed JFSDNet in remote sensing change detection. The reasons are as follows:

1) The frequency-domain information is introduced into the change-detection process to complement the lack of details in mid-level and high-level semantics.
2) The framework can selectively learn the effective ingredients of frequency information to reduce the negative effects of noise.
3) The baseline employs multiple nested encoder–decoder networks to learn more elaborate multiscale features from different levels.

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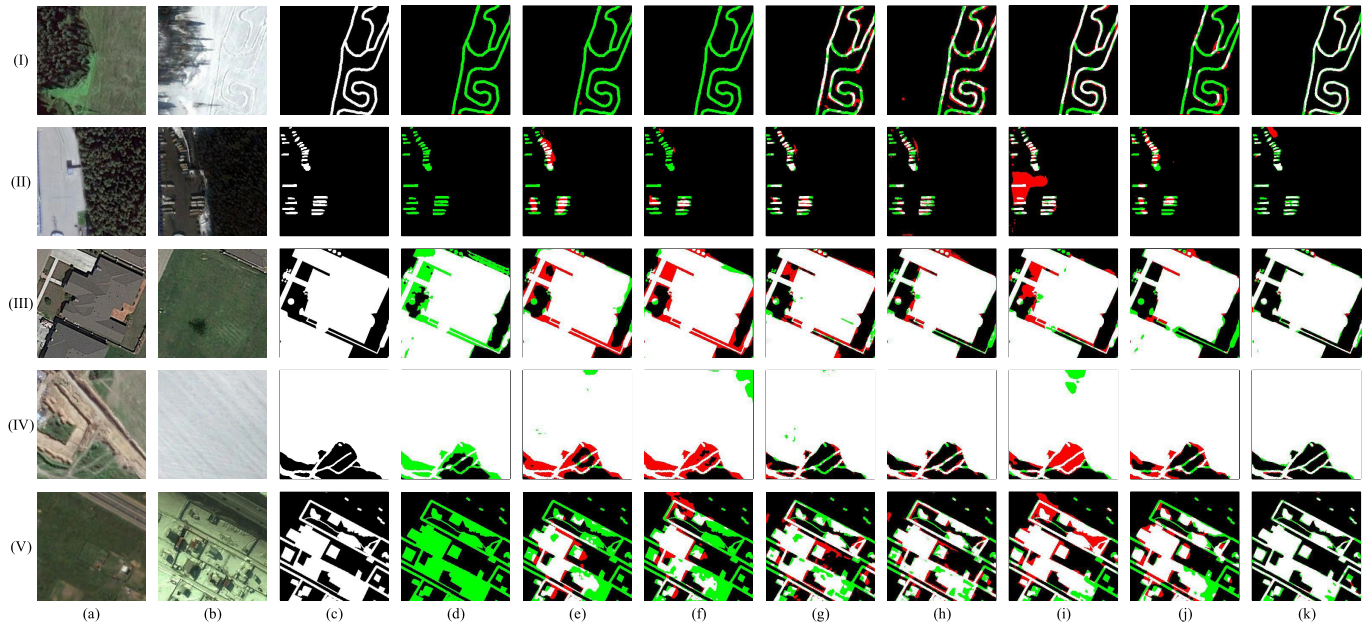


Fig. 5. Change-detection examples of different methods on the CDD dataset. (I) Road change. (II) Minor object change detection. (III) and (IV) Building change. (V) Complex scene change detection. (a) Image1. (b) Image2. (c) Ground truth. Change maps generated by (d) FC-EF, (e) FC-Siam-Conc, (f) FC-Siam-Diff, (g) FCN-PP, (h) Unet++MSOF, (i) DSIFN, and (j) JFSDNet. Different colors are used for a better view: white for true positive, black for true negative, red for FA, and green for MA.

TABLE II
RESULTS OF DIFFERENT APPROACHES ON THE LEVIR DATASET

Method	Pre	Rec	F1	OA	TE	AA	MA	FA	KA
FC-EF (2018, [33])	0.7873	0.7978	0.7872	0.9827	0.0173	0.8922	0.3803	0.0080	0.6799
FC-Siam-Conc (2018, [33])	0.8539	0.8593	0.8544	0.9884	0.0116	0.9412	0.1416	0.0057	0.8613
FC-Siam-Diff (2018, [33])	0.9147	0.8030	0.8528	0.9887	0.0113	0.9411	0.1777	0.0053	0.8385
FCN-PP (2019, [6])	0.9158	0.8417	0.8754	0.9901	0.0099	0.9549	0.1340	0.0042	0.8711
Unet++MSOF (2019, [34])	0.8852	0.8529	0.8669	0.9894	0.0106	0.9394	0.1452	0.0058	0.8869
DSIFN (2020, [41])	0.8561	0.8663	0.8548	0.9897	0.0103	0.9432	0.1628	0.0052	0.8478
BIT-CD (2021, [57])	0.8924	**0.8937**	0.8931	0.9892	0.0108	0.9488	**0.1044**	0.0051	**0.8941**
JFSDNet	**0.9166**	0.8802	**0.8972**	**0.9917**	**0.0083**	**0.9562**	0.1184	**0.0034**	0.8930

B. Comparisons on LEVIR Dataset

1) Quantitative Evaluation: Table II lists the change-detection results for the LEVIR-CD dataset. The JFSDNet obtains excellent results in Pre, Rec, F1, OA, and TE reaching 0.9166, 0.8802, 0.8972, 0.9917, and 0.0083, respectively. BIT-CD has the first-best Rec and the second-best F1, achieving a well-balanced Pre and Rec through transformers, which is lacking in most other methods. Moreover, the metrics of most comparative methods do not have as large a discrepancy as with the CDD dataset, except for the FC-EF being lower than the other methods. The performance improvements of these multiscale designs based on the LEVIR-CD dataset are not as significant as those on the CDD dataset. We presume that this is related to the task of building detection. The change areas of the LEVIR-CD dataset are not as abundant as those of the CDD dataset. The superiority of our network is based on more elusive changes, which have little effect in improving the target for regular changes such as buildings. More details are explained in the qualitative evaluation.

2) Visual Performance: As shown in Fig. 6, the hierarchical fusion-based JFSDNet is visually closer to the ground-truth maps of four samples of bitemporal images from the LEVIR-CD dataset. Some methods can extract main changes, but they perform poorly in several complex scenes (e.g., FC-EF, FC-Siam-Conc, and FC-Siam-Diff). The JFSDNet, FCN-PP, and BIT-CD can locate the incompleteness change areas. In various irregular shape building alterations, the JFSDNET is more accurate than the other approaches (as shown in the first and second rows of Fig. 6). The densely distributed change areas present a challenge, that is, detecting the edges of buildings as completely as possible while maintaining the independence of objects. As shown in the third row of Fig. 6, the JFSDNet can detect objects more accurately and distinguish adjacent buildings by introducing the complement from the frequency domain. However, owing to the effect of shadows, the areas detected by the seven methods have inaccuracies, shown by the unclear contours and incomplete detection of change areas. Like the results on

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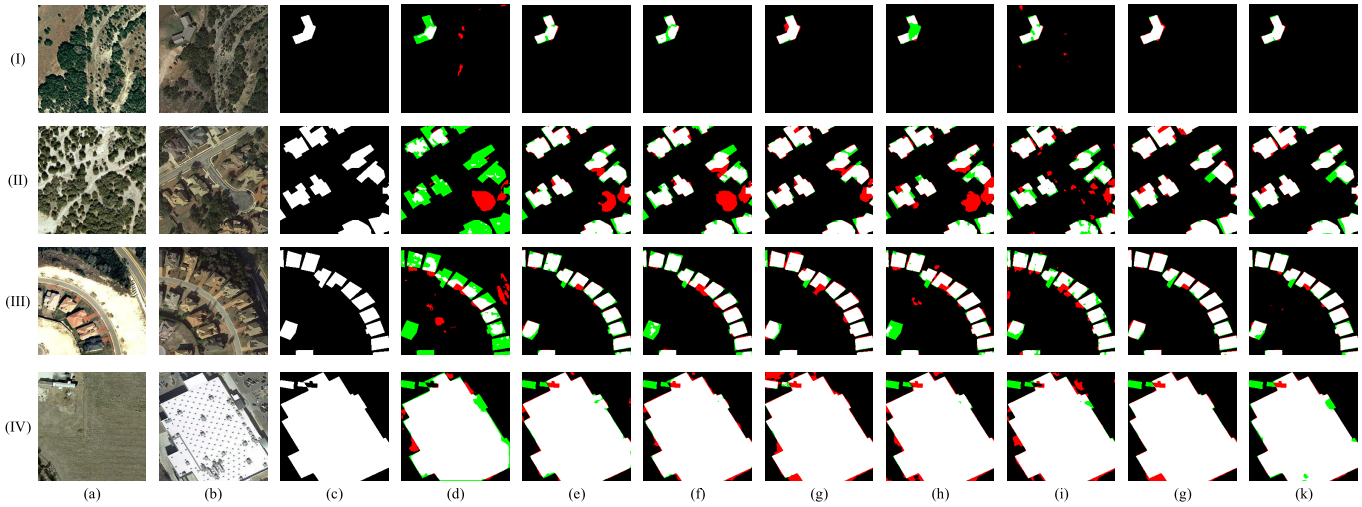


Fig. 6. Change-detection examples of different methods on the LEVIR-CD dataset. (I) and (II) Irregularly shaped buildings change. (III) Densely distributed building change. (IV) Large-scale building change. (a) Image1. (b) Image2. (c) Ground-truth map. Change maps generated by (d) FC-EF, (e) FC-Siam-Conc, (f) FC-Siam-Diff, (g) FCN-PP, (h) Unet++MSOF, (i) DSIFN, (j) BIT-CD, and (k) JFSDNet. Different colors are used for a better view: white for true positive, black for true negative, red for FA, and green for MA.

TABLE III
ABLATION EXPERIMENT OF THE JFSDNET ON THE CDD DATASET

Method	Pre	Rec	F1	OA	TE	AA	MA	FA	KA
FDN	0.9345	0.8858	0.9092	0.9801	0.0199	0.9626	0.1038	0.1037	0.8979
SDN	0.9821	0.8784	0.9269	0.9848	0.0152	0.9845	0.1084	0.0020	0.9270
JFSDNet(without FSM)	**0.9881**	0.9139	0.9493	0.9895	0.0105	0.9896	0.0757	**0.0013**	0.9499
JFSDNet	0.9875	**0.9260**	**0.9555**	**0.9907**	**0.0093**	**0.9899**	**0.0653**	0.0014	**0.9557**

the CDD dataset, the JFSDNet does the best job of keeping complete information of changes among all the baselines. Between all the baselines, our proposed method does the best job of recording all changes, like the results on the CDD dataset.

C. Ablation Study

To verify the validity of both the joint frequency–spatial features and the FSM module, we set the following models for comparison:

1) *Frequency Domain Network (FDN):* the hierarchical reconstruction network in Section III-B that uses only frequency-domain data as its input.
2) *Spatial Domain Network (SDN):* the hierarchical reconstruction model proposed in Section III-B. It takes only spatial-domain data as the input.
3) Our proposed JFSDNet without FSM.

Table III and Figs. 7 and 8 demonstrate the qualitative and quantitative results of the ablation study, respectively. The comparison results in Table III show our JFSDNet outperforms other models on the CDD test sets by a significant margin. The addition of frequency features and FSM have a substantial influence to change detection. The effectiveness of the method using the frequency domain or spatial domain is significantly lower than that of the change-detection method using the joint frequency–spatial domain. The results of our proposed JFSDNet without FSM were significantly better than the SDN: F1 is improved by 0.0224 using the frequency features. After adding the FSM module, the JFSDNet improves F1 by an additional 0.0062 compared to our proposed JFSDNet without FSM. It indicates the importance of the frequency information and the selection of frequency.

Several change-detection examples of different scenes are shown in Figs. 7 and 8. The performance of the FDN and SDN are poor for different changes [as seen in the green and red regions of Figs. 7(d), (e), 8(d) and (e)]. The boundaries of these results are incomplete, especially evident in missing detection problems for small-scale change objects. We can see that the incorporation of frequency-domain information is an effective solution to the misdetection problem. The results in Figs. 7(f) and 8(f) have more explicit boundaries and can detect most of the changes.

The results are further refined based on the above by selective learning, as shown in Figs. 7(g) and 8(g). To summarize, the ablation experiments indicate that the change-detection method with combined frequency- and spatial-domain information can better extract features from the original data. These features complement each other in different levels of the architecture to achieve high-performance change detection. Our proposed JFSDNet can minimize the effects of seasonal changes and different resolutions. It performs well and generates better change maps. However, there are still some weaknesses in the performance of our method (as indicated by the red and green regions of Figs. 7 and 8).

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TABLE IV
EFFICIENCY ANALYSIS ON THE CDD DATASET

Method	FC-Siam-Conc [33]	FCN-PP [6]	Unet++MOSF [34]	DSIFN [41]	Ours
Params. (M)	6.26	13.40	11.00	50.44	11.10
FLOPs (G)	21.69	45.75	45.42	82.27	45.76
F1	0.7131	0.8586	0.8756	0.9030	**0.9555**

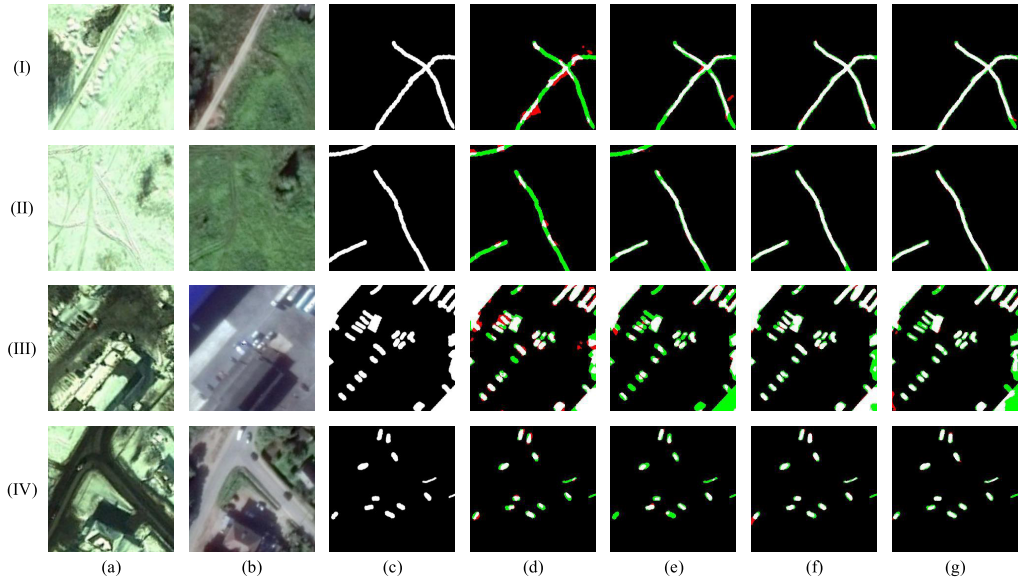


Fig. 7. Ablation study on the CDD dataset. (a) Image1. (b) Image2. (c) Ground truth. Change maps generated by (d) FDN, (e) SDN, (f) JFSDNet without FSM, and (g) JFSDNet. (I) and (II) Road changes. (III) and (IV) Small object changes, such as vehicles. Different colors are used for a better view: white for true positive, black for true negative, red for FA, and green for MA.

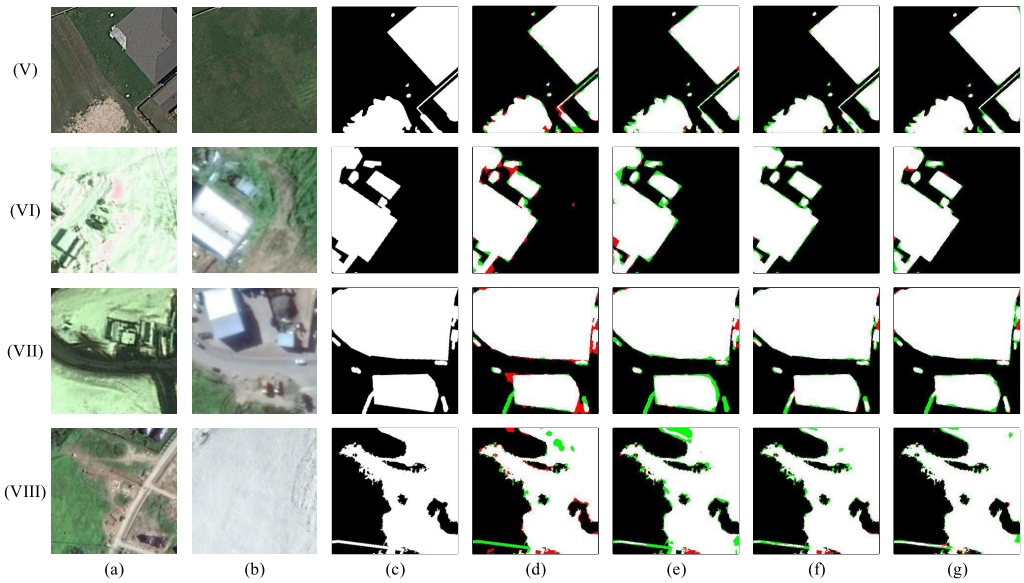


Fig. 8. Ablation study on the CDD dataset. (a) Image1. (b) Image2. (c) Ground truth. Change maps generated by (d) FDN, (e) SDN, (f) JFSDNet without FSM, and (g) JFSDNet. (V) and (VI) Building changes. (VII) and (VIII) Complex changes. Different colors are used for a better view: white for true positive, black for true negative, red for FA, and green for MA.

D. Discussion

1) Efficiency Analysis: We analyzed the effectiveness of our proposed method by comparing it with the four methods covered in the article in terms of a number of parameters (Params.) and floating-point operations per second (FLOPs). To be fair, the input size of all methods is $256 \times 256 \times 3$. As shown in Table IV, among comparison methods, FC-Siam-Conc and FCN-PP have fewer Params and FLOPs. However, due to the

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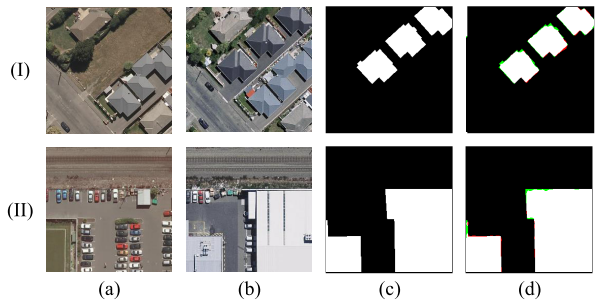


Fig. 9. Case on the postdisaster reconstruction of Christchurch in New Zealand. (a) T1 image. (b) T2 image. (c) Ground truth. Change maps generated by (d) JFSDNet. Different colors are used for a better view: white for true positive, black for true negative, red for FA, and green for MA.

simple design of the network, they are less accurate in change detection. The Unet++MSOF utilizes dense skip connections to learn multiscale features and has medium Params and FLOPs, but its performance is still not satisfactory. The DSIFN employs both transfer learning and multiscale learning to improve performance but leads to a rapid increase in Params (50.44M). Our proposed JFSDNet uses an FSM to filter the frequency-domain data, which can reduce the computational complexity and mine the change clues through the hierarchical network, substantially improving the network performance. Compared with other methods, the JFSDNet achieves high performance with an affordable parameter number and FLOPs, having a better tradeoff between model size and performance.

2) Case Analysis: To demonstrate the advantage of our proposed approach, we select the postdisaster reconstruction after the 6.3 magnitude earthquake in Christchurch, New Zealand, as a case. The bitemporal images are obtained in the same area in April 2012 and 2016, respectively. The number of buildings in the area increased significantly during the four years. We chose two areas of size 256*256 for building change analysis (as Fig. 9 shown). The first row in Fig. 9 shows the addition of some new buildings to the original building stock in the area. Our JFSDNet performs well in detecting the increase of buildings in this area. The second row in Fig. 9 shows the change in large buildings. Our network can accurately detect changes from parking lots to large buildings. Our proposed avoids the interference of cars and roads, which can detect the target change area more accurately. Specifically, the change map detected by JFSDNet is almost the same as the ground truth. The analysis above indicates that our frequency-domain branch can provide the neural network with edge information of the object to distinguish between similarly colored areas.

VI. Conclusion

In this article, an innovative framework with joint spatial–frequency-domain learning is proposed for remote sensing optical image change detection. The changed features in the frequency domain are effectively introduced to the framework and used to adaptively complement the absence of deep-level features. In addition, we design a frequency-domain selection module to reduce the complexity of the frequency-domain information and use it more effectively. In the training process, supervising multiple outputs allowed the hierarchical neural network to receive constraints. Results of compared experiment and ablation study on the CDD dataset and the LEVIR-CD dataset demonstrate that the proposed strategy has better detection performance than the other compared methods, especially for complex change detection of high-resolution remote sensing optical images. In the future, our research will be conducted on remote sensing images from different data sources and fully mine the association between spatial-domain and frequency-domain features.

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